When Trading Dries Up

A liquidity signal for a long-only fund

Group 7
Silver Fund

Built on 2011–2018

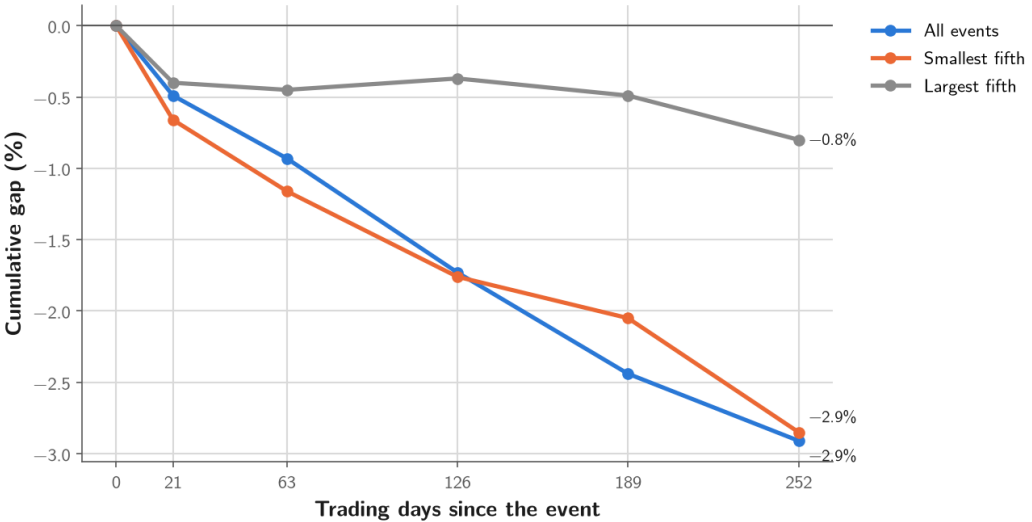
The idea

- A **liquidity event**: a stock suddenly trades far less than usual for it (measured against its own past year).
- **Why it should matter:** most investors avoid illiquid stocks; only a small group of patient investors holds them, at a slightly lower price.
- When a stock's trading collapses, it has to pass from its current holders to that small group.
  - Sellers can only exit slowly, because volume has dried up; the few buyers need a lower price.
  - So the price **walks down for months**. Shorting these stocks is costly, so nobody trades the drift away.
- We first remove anything the fund's six existing scores already capture.
- Questions: is the effect real, and can our long-only fund make money from it?

Bottom line

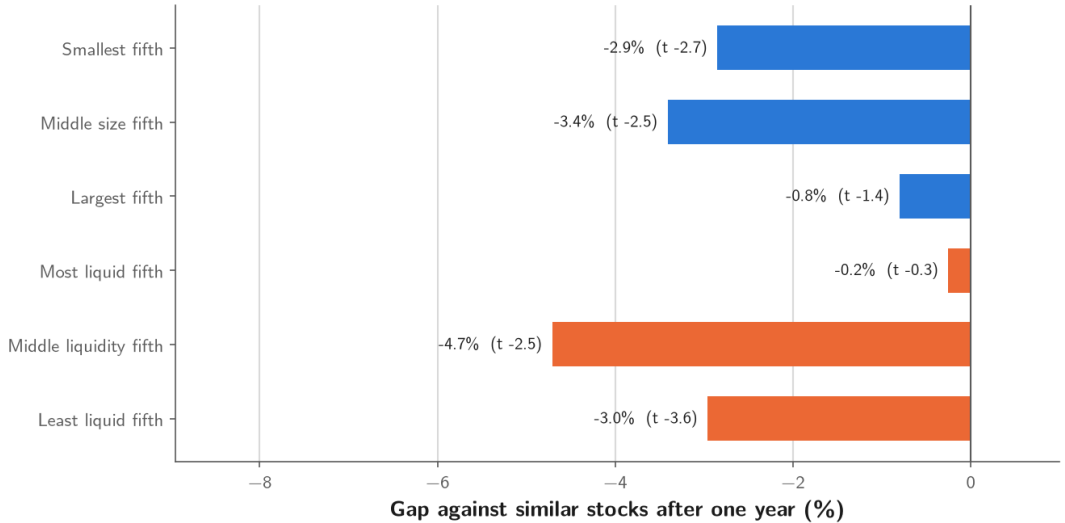
- **Yes, it is real.** Stocks whose trading dries up keep falling: about 3% behind similar stocks over the next year.
- **Just avoiding them doesn't help:** $+0.05\%$ to $+0.10\%$ a year, not significant.
- **Betting against them daily with the optimizer works:** long-only after trading costs $+0.12\%$ a year (IR 1.04).
- **It is small.** Use it as one input to the fund's alpha.

After trading dries up, the stock keeps falling



Return after a liquidity event minus similar stocks (same size and liquidity) on the same day; 2006–2018. All events: t = -3.6 at one year.

Mostly in small, illiquid stocks



Gap against similar stocks one year after the event, by size (blue) and by how illiquid the stock was before (orange); 2006–2018.

It is not something we already know

	How much less flagged stocks earn, %/month (t)
Had a liquidity event in the last month	0.18 (2.5)
... also controlling for last month's return	0.17 (2.4)

Positive = flagged stocks earn less, which is what we bet on. Monthly regressions of next month's return, 2008–2018; every row also controls for size, how illiquid the stock already was, its past-year return and its return two to three years ago, and the fund's six scores were removed before flagging. Measured once a month, so it understates the first-weeks effect that daily trading captures.

The simple idea doesn't work: just avoid these stocks

2011–2018	Weight avoided	Return	Risk	IR (t)	IR, market removed
Just avoid flagged stocks, daily	8.6%	+0.05%/yr	0.47%	0.11 (0.3)	−0.12
Just avoid flagged stocks, monthly	8.7%	+0.10%/yr	0.43%	0.23 (0.7)	−0.12
Optimizer, daily, no shorting, costs	—	+0.12%/yr	0.11%	1.04 (2.9)	—

- Hold the benchmark, sell every stock with a liquidity event in the last 63 days, give its weight to similar stocks.
- It barely earns anything: these stocks are small, so dropping them tilts the portfolio in ways that add noise and swamp the effect.
- The optimizer removes that noise and sizes each bet. That is what makes the signal pay.

How we bet on it

Our alpha (Grinold–Kahn form): $\alpha_i = \sigma_i \times \bar{d}_{\text{age of event}}$

- σ_i: volatility; $\bar{d}$: how far flagged stocks fell behind similar stocks per unit of volatility (ages 1–21, 22–63 days), from earlier years. No recent event: $\alpha = 0$.

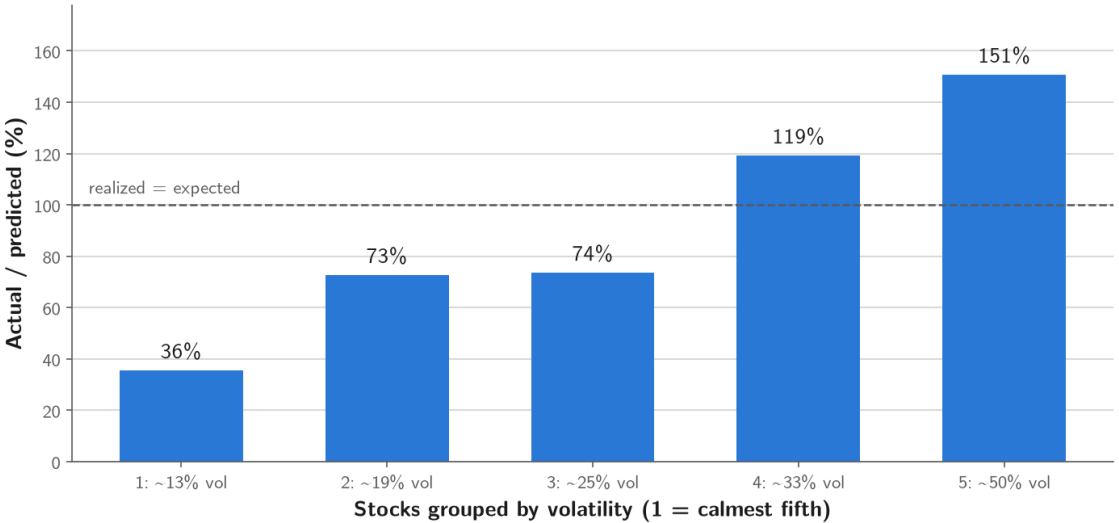
Optimizer: maximize $h^\top \alpha - \frac{\gamma}{2} h^\top V h - \lambda \cdot$ spread costs, active risk $\leq 1\%$

- γ is fixed for a year: the previous year's median of the value that would put the no-limits portfolio at 1% risk.
- λ (0.5–4) is picked each January from earlier years' net results.

Constraint	No limits	Limited shorting + costs	No shorting + costs
Fully invested: active weights sum to zero	✓	✓	✓
Zero beta to the benchmark	✓	✓	✓
Net zero within each size × liquidity group	✓	✓	✓
Short at most 0.25% per stock ($b + h \geq -0.25\%$)	–	✓	–
Long-only: no weight below zero ($b + h \geq 0$)	–	–	✓
Pays half the bid–ask spread on every trade	–	✓	✓

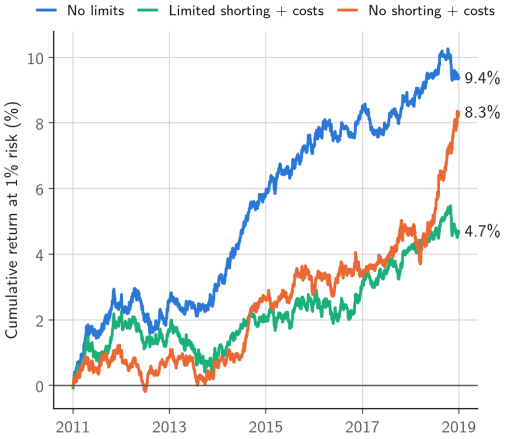
No limits pays no costs: it is the yardstick for the signal's full strength. No shorting stays small: the stocks it bets against hold little benchmark weight.

Calm stocks fell less than we expected, volatile stocks more



An earlier version left volatility out of the Grinold–Kahn alpha. Bars: actual return as % of predicted, 2011–2018. So alpha now scales with volatility.

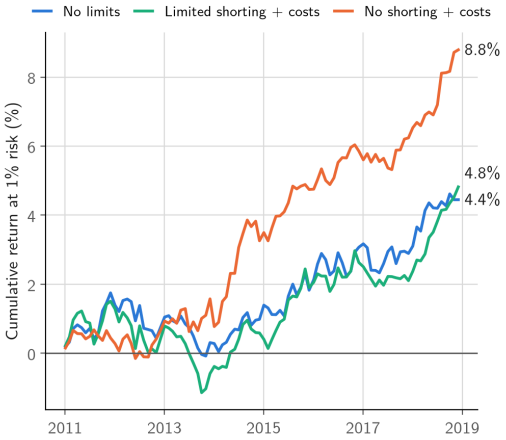
Daily rebalancing, 2011–2018



	No limits	Limited shorting + costs	No shorting + costs
Active return, %/yr	+1.30	+0.57	+0.12
Active risk (realized)	1.11%	0.97%	0.11%
IR (t)	1.17 (3.3)	0.59 (1.7)	1.04 (2.9)
Max drawdown	−1.53%	−1.79%	−0.16%
Positive years	8/8	6/8	7/8
Turnover, ×/yr	35.1	11.6	0.8
TC vs no limits	ref.	0.84	0.33
Spread cost, %/yr	2.64	0.52	0.01

Each period's line scaled to 1% realized risk; the table shows actual numbers. No shorting = total weight ≥ 0. Limited shorting = total weight $\geq -0.25\%$. + costs = pays half the spread on each trade; returns are after those costs. Fixed γ, risk $\leq 1\%$. TC = ex-ante transfer coefficient vs. no limits.

Monthly rebalancing, 2011–2018



	No limits	Limited shorting + costs	No shorting + costs
Active return, %/yr	+0.62	+0.43	+0.14
Active risk (realized)	1.11%	0.71%	0.12%
IR (t)	0.56 (1.6)	0.60 (1.7)	1.10 (3.1)
Max drawdown	−2.03%	−1.89%	−0.10%
Positive years	5/8	5/8	8/8
Turnover, ×/yr	11.3	2.9	0.3
TC vs no limits	ref.	0.58	0.24
Spread cost, %/yr	1.04	0.13	0.01

Each period's line scaled to 1% realized risk; the table shows actual numbers. No shorting = total weight ≥ 0. Limited shorting = total weight $\geq -0.25\%$. + costs = pays half the spread on each trade; returns are after those costs. Fixed γ, risk $\leq 1\%$. TC = ex-ante transfer coefficient vs. no limits.

How much to trust it

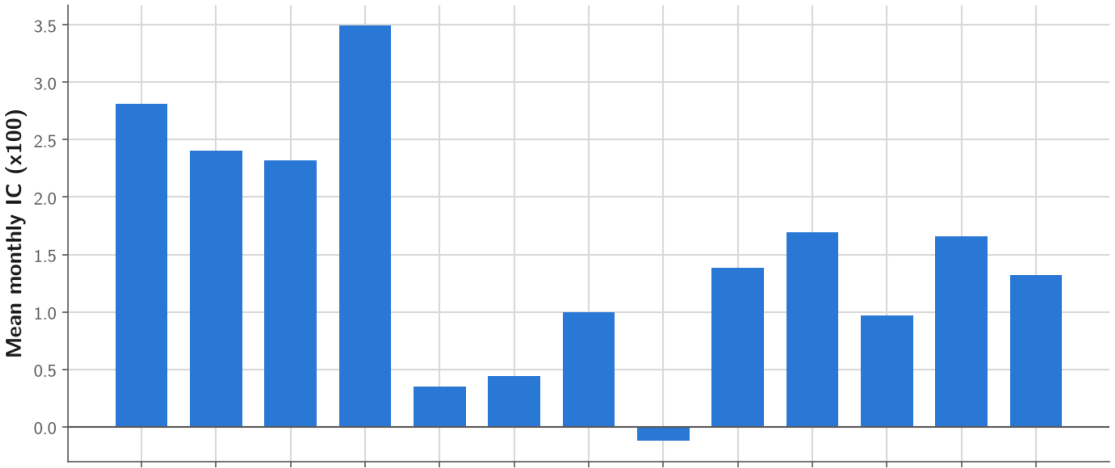
- We built it on 2011–18 in four rounds; 2019–24 is held back for a final test.

Conclusion

- Stocks whose trading dries up keep falling for months: they must pass to a small group of patient buyers, and that hand-off is slow.
- A long-only fund can make money from it, but only with the optimizer and daily trading.
- It is small. Use it as one input to the fund's alpha, not on its own.
- Next: combine it with the fund's scores and test the combined alpha, with trading costs.

Appendix

Appendix: new information beyond the fund's scores? IC by year



Rank IC with next-month Barra specific return, fund's six scores removed. Mean 0.015 (t 6.0); 12 of 13 years positive.

Appendix: what did we change, and when?

Daily runs, %/yr (IR), no costs	No limits	No shorting
1. Flag that turns over daily	−0.9 (−0.15)	—
2. Event clock	−2.77 (−0.49)	−0.12 (−0.24)
3. + no positive alpha, hedge with similar stocks	+3.23 (0.58)	+0.36 (1.48)
4. + alpha scaled by volatility	+6.84 (1.17)	+0.19 (1.82)

- Version 2: a flag that switched off after a few days missed most of the drift, so we gave each event an age.
- Version 3: a noisy positive alpha for large stocks made the portfolio buy them; we stopped allowing positive alpha and hedged with similar stocks.
- Version 4: calm stocks fell less than predicted (slide 9), so alpha now scales with volatility. Frozen before the 2019–24 test.

Appendix: how we picked the short limit (2011–2018)

Short limit per stock	Active risk (model)	IR	Turnover, ×/yr	Monthly IR
0.05%	0.47%	1.14	15	0.74
0.1%	0.85%	1.02	27	0.65
0.25%	**1.68%**	**0.97**	**55**	**0.54**
0.5%	2.65%	0.94	90	0.54
1%	3.90%	1.03	139	0.48
2%	4.71%	1.12	175	0.50

Daily runs unless marked, earlier design (5% risk target). The IRs are within noise of each other, so we chose the limit by risk: 0.25% (bold) sits between long-only and no limits. The same limit is used monthly.

Appendix: every portfolio with and without trading costs (2011–2018)

		Return, %/yr	IR (t)	Risk	Turnover, ×/yr	Spread cost, %/yr
Daily rebalancing						
No limits	no costs	+1.30	1.17 (3.3)	1.11%	35	2.64*
No limits	with costs	+0.74	0.71 (2.0)	1.05%	14	0.58
Limited shorting (0.25%)	no costs	+1.09	1.07 (3.0)	1.02%	30	2.34*
Limited shorting (0.25%)	with costs	+0.57	0.59 (1.7)	0.97%	12	0.52
No shorting	no costs	+0.16	1.97 (5.6)	0.08%	2	0.08*
No shorting	with costs	+0.12	1.04 (2.9)	0.11%	1	0.01
Monthly rebalancing						
No limits	no costs	+0.62	0.56 (1.6)	1.11%	11	1.04*
No limits	with costs	+0.38	0.51 (1.4)	0.76%	3	0.13
Limited shorting (0.25%)	no costs	+0.53	0.52 (1.5)	1.02%	9	0.89*
Limited shorting (0.25%)	with costs	+0.43	0.60 (1.7)	0.71%	3	0.13
No shorting	no costs	+0.07	0.98 (2.8)	0.07%	1	0.02*
No shorting	with costs	+0.14	1.10 (3.1)	0.12%	0	0.01

Fixed γ, active risk $\leq 1\%$. Costs = half the bid–ask spread on every trade, charged in the optimizer and in the returns; * = what the portfolio would pay if traded alone (not charged). Risk = realized.

Appendix: where does the return come from? (2011–2018)

Return, %/yr	Event 1–21 days ago	Event 22–504 days ago	No recent event
Daily, no limits	+14.88	−3.43	−4.64
Daily, no shorting	+0.12	+0.08	−0.01
Monthly, no limits	+8.02	+0.09	−4.09
Monthly, no shorting	+0.09	+0.06	−0.05

Return earned on positions in stocks with a recent event, an older event, or none. In the no-limits runs the bets against flagged stocks are hedged with similar stocks; small, illiquid stocks lag in general, so both sides show it and it cancels.